

Cyber Security

Respond to a teacher-provided case study to examine a cyber security incident or network vulnerability, evaluate threats, and propose strategies to protect data and information on the network.

The big picture

This outcome is built around a case study. You first understand the network — its hardware and software, the types of network (wired, wireless, mobile; LAN, WAN, WPAN; intranet, internet, VPN) — then analyse the threats (malware, denial-of-service, weak credentials, outdated software) and the risks of operating on global networks. From there you propose protections: multi-factor authentication, backups, updates, staff procedures, cryptography and frameworks such as the Essential Eight. AI's role in detection and the place of ethical hacking and legislation round it out.

How it's assessed

Case-study analysis is the core skill: read the scenario, identify the specific threat or vulnerability, and propose a protection that actually addresses it. Expect to distinguish malware types, explain how a threat works, recommend and justify risk-reduction measures, explain cryptography (symmetric vs asymmetric, a named cipher), and name relevant legislation or the Essential Eight. Listing generic "use antivirus" advice without matching it to the scenario's threat loses marks.

Key knowledge at a glance

KK	Topic	In one line
KK01	Emerging trends in cyber security	The walls that used to protect a network have basically dissolved. This dot point is about the five ideas the whole industry is shifting toward — and why "keep the bad guys outside" is no longer a plan.
KK02	The hardware & software that hold a network together	Every cyber security incident in this course happens on a network. Before you can defend one, you need to know the gear inside it — what each box does, and which ones are actually there to keep attackers out.
KK03	Wired, wireless & mobile networks	Every network has to make the same bargain: speed and safety, or freedom to move. This module shows you how to weigh wired, wireless and mobile against the four things examiners care about — and how to justify a choice instead of just naming one.

KK	Topic	In one line
KK04	Three networks, one question: who do you trust, and how far can you reach?	The internet is the worldwide, public system that connects billions of independently-owned networks so they can exchange data using one shared set of rules.
KK05	LANs, WANs & WPANs	Three letters in front of "AN" decide how far a network reaches, how fast it runs, who owns it, and how exposed your data is. This dot point is about matching the right network type to the right job — and being able to defend that choice in an exam.
KK06	Risks & benefits of networks in a global environment	Above: PixelPort, run from a bedroom in Melbourne, with players, servers and a payment processor scattered across the globe. Spin the planet and switch on each layer — every line is something that has to stay safe and legal in more than one country at once.
KK07	Malware	The technical underpinnings of spyware, viruses, worms and ransomware — what each one actually is, how it gets in, and how it behaves once it's there.
KK08	Security threats	A security threat is any potential cause of an unwanted event that could harm a network, its data, or the people who rely on it. This dot point names five of them. Your job in the exam is not just to define each one — it's to recognise it inside a case study and explain the damage it does. Let's get the vocabulary straight.
KK09	Reducing the risk — five ways to keep a network harder to break.	You can't make a network unhackable. But you can make it expensive, slow and frustrating to attack — and quick to recover when something slips through. That's what risk reduction means.
KK10	Cryptography: turning a readable note into unreadable noise	Anything you send across a network can be copied by someone in the middle. Cryptography doesn't stop them grabbing it — it makes what they grab useless. This module walks the whole story, from a 2,000-year-old shift cipher to the encryption protecting your bank login right now.
KK11	AI in Network Security	The VCAA study design asks you to understand the use of artificial intelligence in network security — specifically machine-learning analysis of traffic patterns and real-time monitoring and notification. Two ideas, one job: spot trouble fast.
KK12	Ethical hacking Breaking in — with permission — so the criminals can't.	Ethical hacking means using the same tools and techniques a criminal would — but with the owner's written permission, and the goal of fixing weaknesses instead of exploiting them.
KK13	Legislation & Frameworks	The laws every Australian network must obey, and the frameworks that turn "be secure" into a real to-do list. We map all five — through the eyes of a Melbourne learner-driver app called Clutch.

Chapter glossary

Term	Meaning
Malware	Malicious software, including viruses, worms, spyware and ransomware.
Ransomware	Malware that encrypts files and demands payment for their release.
Denial-of-service (DoS)	An attack that floods a system to make it unavailable to legitimate users.
Multi-factor authentication	Verifying identity with two or more independent factors.
Firewall	Hardware/software that filters network traffic against security rules.
Symmetric encryption	Encryption using one shared key to both encrypt and decrypt (e.g. AES).
Asymmetric encryption	Encryption using a public/private key pair (e.g. RSA, ECDH).
VPN	A virtual private network providing a secure, encrypted connection over a public network.
Essential Eight	A baseline set of mitigation strategies for reducing cyber security risk.

Answering across the chapter

These scenario questions pull together several key-knowledge points at once — the kind of integrated thinking the exam rewards. Compare the weak and strong responses.

Q1. Analyse · 4 marks

A small business is hit by ransomware after an employee opens an email attachment. Analyse how the attack succeeded and recommend two protections, justifying each.

✗ A weak answer

They got a virus from email and should get antivirus and be careful. (→ *Wrong category (ransomware ≠ generic virus), one vague control — 1.*)

✓ A strong answer

The **ransomware** reached the network through a phishing email and encrypted the business's files, exploiting a human entry point and likely unpatched software. Protection one: **regular offline backups**, so the business can restore data and refuse the ransom — removing the attacker's leverage. Protection two: **staff training plus email filtering and patching**, which reduces the chance the attachment is opened and closes the vulnerabilities ransomware uses. Each directly addresses how this attack succeeded.

How the marks are awarded

- 1 mark — correctly identifies ransomware and the phishing/human entry point.
- 1 mark — a valid protection (offline backups) with justification.
- 1 mark — a second valid protection (training/filtering/patching).
- 1 mark — protections are tied to how the attack succeeded.

Q2. Explain · 3 marks

Explain the difference between symmetric and asymmetric encryption and give one situation suited to each.

✗ A weak answer

Symmetric uses one key and asymmetric uses two keys. (→ *Correct fact but no situations, so it caps at 1-2.*)

✓ A strong answer

Symmetric encryption uses a single shared key to encrypt and decrypt — it is fast, so it suits encrypting large stored files (e.g. AES on a backup drive). **Asymmetric** encryption uses a public/private key pair — the public key encrypts and only the matching private key decrypts — which suits securely exchanging data with someone you have not shared a key with, such as setting up an HTTPS connection. The trade-off is speed versus key-sharing.

How the marks are awarded

- 1 mark — symmetric = one shared key.
- 1 mark — asymmetric = public/private key pair.
- 1 mark — a suitable situation for each.

Chapter revision checklist

- Describe network hardware/software and the types of network (LAN/WAN/WPAN, intranet/internet/VPN).
- Compare wired, wireless and mobile networks on cost, reliability and security.
- Distinguish malware types and explain how each works.
- Identify security threats and global-network risks in a case study.
- Recommend and justify risk-reduction measures (MFA, backups, updates, procedures).
- Explain symmetric vs asymmetric encryption and a named cipher.
- Describe AI's role in detection and the role of ethical hacking.
- Name relevant legislation and the Essential Eight.